

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of MSc (MI) Examination March 2019

Course code & Name: MI 821 Industrial Microbiology

Date: 14/3/2019 Day: Thursday Time: 1:30 pm To 2:00 pm Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions.

20

- Classical fermentation and bioprocess technology is often known as ____ technology.
(i) Red (ii) Green
(iii) White (iv) Gray
- Starch is hydrolysed by the following enzyme
(i) amylase (ii) protease
(iii) lipase (iv) nuclease
- _____ is the byproduct of biodiesel production process.
(i) Glyceric acid (ii) Glycerol
(iii) Acetic acid (iv) Acetone
- The following ions is a precursor For the production of vitamin B12 (cyanocobaltamine)
(i) cobalt (ii) chromium
(iii) calcium (iv) potassium
- Phenylacetic acid is added as precursor during production of ____
(i) Penicillin G (ii) Penicillin V
(iii) Cephalosporin (iv) Cephamycin
- Rhodotorula sp. produces pigments using _____ as a substrate.
(i) Sugarcane (ii) Agricultural wastes
(iii) Apple pomace (iv) None of the above

P.T.O

7. The following proteolytic enzyme is used in detergents as they function at high pH
 - (i) trypsin
 - (ii) chymotrypsin
 - (iii) subtilisins
 - (iv) all of the above
8. EDTA inhibits the following proteases by chelating metal ions
 - (i) Serine
 - (ii) Aspartyl
 - (iii) Metallo
 - (iv) None of the above
9. _____ is unicellular algae used as protein rich biomass.
 - (i) Spirulina
 - (ii) Chlorella
 - (iii) Senedasmus
 - (iv) Both, (ii) and (iii)
10. The concentration of Mn^{+2} and Mg^{+2} should be kept low in the fermentation medium for the production of the following
 - (i) lactic acid
 - (ii) riboflavin
 - (iii) penicillin
 - (iv) citric acid
11. The following may be a part of quality control system
 - (i) Raw material procurement
 - (ii) testing of finished product
 - (iii) Proper waste disposal system
 - (iv) All of the above
12. The following action can be taken by a technician if a drug fails the QC
 - (i) Order reprocessing of the drug
 - (ii) Reject the drug
 - (iii) Report it as fail
 - (iv) Recall the product
13. The following organism has maximum tendency to move toward a state of lower productivity when used for a fermentation
 - (i) wild type
 - (ii) mutant
 - (iii) natural variant
 - (iv) all of the above
14. Food containing *Bifidobacterium longum* is an example of the following
 - (i) probiotic
 - (ii) prebiotic
 - (iii) postbiotic
 - (iv) synbiotic
15. PRM/PIM or BRM/BIM is used to inhibit the proliferation of the following in a dairy starter
 - (i) microbial contaminants
 - (ii) starter microbe
 - (iii) bacteriophage
 - (iv) all of the above
16. Use of defined medium in a fermentation process increases the cost of the following
 - (i) downstream processing
 - (ii) intermediate fermentation
 - (iii) fermentation process
 - (iv) inoculum generation
17. The microbes involved in bioleaching of metal derive their energy from the following
 - (i) glucose
 - (ii) metal oxidation
 - (iii) metal reduction
 - (iv) CO_2
18. A following is mandatory for organic synthesis using microorganisms or enzymes
 - (i) chiral C
 - (ii) C forming aromatic ring
 - (iii) chiral H
 - (iv) C forming aliphatic chain
19. Blue triazine dye used in Blue SEPHAROSE Affinity Chromatography **removes** the following
 - (i) chaperones and foldases
 - (ii) disulfite bonds
 - (iii) proteolytic enzymes
 - (iv) kinases
20. The recombinant insulin produced as a fusion protein is treated chemically with the following to yield proinsulin
 - (i) Acetone
 - (ii) gluteraldehyde
 - (iii) cyanogen bromide
 - (iv) potassium bromide

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III Semester of MSc (MI) Examination March 2019

Course code & Name: MI 821 Industrial Microbiology

Date: 14/3/2019 Day: Thursday Time: 2:00 pm To 4:30 pm Maximum Marks: 50

Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEET.
 2. Make suitable assumptions and draw neat figures wherever required.
 3. Use of non-programmable calculator is allowed.
 4. Show necessary calculations.
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SECTION - I

20

Q – II A Discuss briefly the downstream processing of insulin produced using recombinant *E.coli* K12 (04)

OR

Discuss briefly: the production of human interferon by the cultivation of human cells

Q – II B Discuss briefly any two of the following: (06)

- (i) The criteria to be considered for inoculum generation
- (ii) The criteria to be considered for validation of fermentation process
- (iii) Tiered seed banking system

Q – II C Attempt any two of the following (04)

- (i) Conversion of Penicillin G to Ampicilin as an example of semisynthetic antibiotic
- (ii) Discuss the importance of QA and QC in pharmaceutical industries
- (iii) Write a short note on CGMP

Q – II D Discuss briefly any two of the following (06)

- (i) Heap bioleaching of copper-containing ore
- (ii) Strategies used in screening of biocatalyst for biotransformations
- (iii) Biotransformations catalyzed by lipases/esterases

SECTION - II

30

Q – IIIA Write a comprehensive note on any one of the following (05)

- (i) Short note: Different generations of antibiotics
- (ii) Short note: Different techniques used in enzyme immobilization

Q – IIIB Justify the statement: Pen. G cannot be given orally but Pen. V can. (02)

OR

Highlight advantages and limitations associated with the use of SCPs?

P.T.O

- Q – IVA Attempt any six of the following (18)**
- (i)** Explain consolidated bioprocessing in reference to bioethanol fermentation.
 - (ii)** Discuss briefly downstream processing of L-lysine fermentation.
 - (iii)** Explain in brief: types of proteases
 - (iv)** Which are the factors that may affect pigment production?
 - (v)** How rate of aeration can affect citric acid fermentation?
 - (vi)** Differentiate between alpha-, beta- and glucoamylases
 - (vii)** What are different types of patents?
- Q – V A Discuss briefly any two of the following (04)**
- (i)** Mechanically protected systems for the production of starter cultures
 - (ii)** Characteristics of microbes used as Probiotics
 - (iii)** Types of bioinoculants
- Q – V B Justify: Acidophiles with metal ions resistance are involved in biomining (01)**